

Voyana

AI-Powered Collaborative Travel Ecosystem

Project Title: Voyana – AI-Powered Collaborative Travel Ecosystem

Course: Software Engineering — Semester Project

Team Size: 8 Members

Team Members

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2. Short Description

Voyana is a collaborative travel platform that brings trip planning, bookings, budgeting, group communication, and travel memories into a single ecosystem instead of the usual scatter of apps, spreadsheets, and chat threads. Built on a Service-Oriented Architecture, the platform is organized as a set of independent services — authentication, trip planning, bookings, expense management, AI assistance, and more — that communicate through REST APIs and can be developed, tested, and scaled on their own. Artificial intelligence is woven through the platform rather than confined to a single chatbot: it drafts itineraries, recommends destinations, tracks budgets, and turns a finished trip into a shareable journal. The result is a system that lets a group of travelers plan together, split costs fairly, and hold on to the memories afterward, all from one place.

3. Application Type

Voyana is delivered as a responsive web platform rather than a native mobile app, which keeps the scope manageable for the team while still covering both traveler-facing and admin-facing needs:

- **Responsive Web Application (Traveler & Admin)** — a single React.js application, built mobile-responsive with Tailwind CSS, that adapts to desktop, tablet, and phone browsers. Travelers use it for trip planning, group chat, bookings, and expense tracking, while administrators access role-gated admin views within the same application, removing the need for a separate native mobile app.
- **REST API Services** — the backend is exposed as a set of independently deployable REST services, documented with Swagger/OpenAPI, so the web application and any future client can consume the same endpoints.
- **Cloud-Based Architecture** — services are containerized with Docker and deployed on Google Cloud Platform, which gives the team independent scaling for high-traffic services (such as AI itinerary generation) without over-provisioning lighter services (such as document storage).

4. Motivation / Problem Statement

Planning a trip today, especially a group trip, usually means juggling several disconnected tools: a flight booking site, a hotel app, a spreadsheet for expenses, a group chat for logistics, and a separate app or camera roll for photos afterward. This fragmentation creates several recurring problems for travelers:

- Using multiple disconnected applications for planning, booking, and payments, with no single source of truth for the trip.
- Poor collaboration among travel groups — decisions on dates, destinations, and activities happen over scattered messages instead of a shared plan.
- Difficult expense management, where tracking who paid for what and settling up afterward is manual and error-prone.
- Time-consuming planning, since building a day-by-day itinerary from scratch takes hours of research.
- Lack of centralized travel information — weather, currency rates, visa requirements, and local transport options are spread across different sources.
- No intelligent travel assistant that can answer trip-specific questions or adapt plans on the fly.
- Fragmented bookings across airlines, hotels, and local transport providers with no unified view of confirmations.
- Difficulty preserving travel memories in a way that's tied to the actual itinerary rather than a generic photo gallery.

Voyana addresses these problems by consolidating planning, booking, budgeting, communication, and memory-keeping into one platform, with AI assistance available at each stage of the trip rather than as a bolt-on feature. Because the platform is built as a set of independent services, new capabilities — a new booking partner, a new AI feature — can be added without reworking the rest of the system.

5. Target Users

Voyana is designed to serve a range of user groups, each with different needs from the platform:

User Group	Description	How They Benefit
Solo Travelers	Individuals planning and taking trips independently.	AI-generated itineraries and recommendations replace hours of manual research.
Friends	Small groups planning leisure trips together.	Shared itineraries, polls, and expense splitting remove the usual back-and-forth.
Families	Multi-generational groups with varied needs and schedules.	Shared calendars and task assignments keep everyone aligned on plans.
Students	Budget-conscious travelers, often planning group trips.	Smart budget planning and settlement tracking help manage limited funds fairly.
Corporate Teams	Employees traveling for business or offsites.	Centralized bookings and document storage simplify trip logistics and reporting.
Tour Organizers	Individuals coordinating trips for larger groups.	Task assignment, group chat, and voting features streamline coordination.
Travel Agencies	Businesses that plan trips on behalf of clients.	Admin dashboard and analytics support managing multiple client trips at once.
Backpackers	Independent, often long-term or multi-destination travelers.	Local transport info, currency conversion, and emergency assistance support travel on the go.
Administrators	Platform staff managing users and content.	Admin Management and Analytics services provide oversight and moderation tools.

6. Use Cases

The table below summarizes the core use cases supported by Voyana, grouped roughly by the stage of the travel journey they support.

#	Use Case	Primary Actor	Description
1	User Registration	Traveler	Create an account with email/social sign-up and set up a traveler profile.
2	Login	Traveler	Authenticate via JWT-based login, including OAuth options.
3	Create Trip	Trip Organizer	Start a new trip with destination, dates, and budget range.
4	Invite Members	Trip Organizer	Add co-travelers to a trip via link, email, or username.
5	Join Trip	Traveler	Accept an invitation and gain access to the shared trip workspace.
6	AI Generates Itinerary	AI Travel Assistant	Produce a draft day-by-day itinerary based on preferences and constraints.
7	Edit Itinerary Collaboratively	Trip Members	Multiple members adjust the shared itinerary in real time.
8	Vote on Destination/Activity	Trip Members	Group members vote on options proposed for the trip.
9	Book Flights	Traveler	Search and reserve flights through the Flight Booking Service.

#	Use Case	Primary Actor	Description
10	Reserve Hotels	Traveler	Search and book accommodation through the Hotel Booking Service.
11	Manage Local Transport	Traveler	View and arrange local transport options at the destination.
12	Split Expenses	Trip Members	Log shared expenses and divide costs among members.
13	Track Settlements	Trip Members	View who owes whom and record repayments.
14	Make Payments	Traveler	Pay for bookings or settle shared expenses via the Payment Service.
15	Chat with AI Assistant	Traveler	Ask trip-specific questions and get contextual answers.
16	Store Travel Documents	Traveler	Upload and securely store passports, visas, and tickets.
17	Upload Travel Memories	Traveler	Add photos and notes to a trip's memory album.
18	Generate Travel Journal	AI Travel Assistant	Compile a trip summary and journal from itinerary and memories.
19	Receive Notifications	Traveler	Get alerts for booking updates, itinerary changes, and reminders.
20	Leave Reviews	Traveler	Rate and review destinations, stays, and activities.
21	View Analytics	Admin / Organizer	View spending, engagement, and trip statistics.
22	Manage Users	Admin	Suspend, verify, or manage traveler and partner accounts.

7. Scope & High-Level Features

Features are organized into modules that map directly onto the platform's services, reflecting the Service-Oriented Architecture described in Section 8.

User Management

- Traveler profile creation and editing
- Profile picture and travel preference settings
- Friend/connection list for frequent travel companions
- Account deactivation and data export
- Preference-based personalization settings

Authentication

- Email/password registration with verification
- OAuth login (Google, Apple)
- JWT-based session management
- Role-based access control (traveler, organizer, admin)
- Password reset and multi-device session handling

AI Services

- AI-generated day-by-day itineraries
- Personalized destination recommendations
- Smart budget planning based on trip parameters

- Route optimization for multi-stop itineraries
- Conversational AI travel assistant
- Weather-aware activity suggestions
- Restaurant and activity recommendations
- AI-generated packing checklists
- Visa and travel guideline assistance
- Trip summary generation
- AI-generated travel journals
- AI-generated photo captions for memories
- Intelligent reminders (check-in times, document expiry, etc.)

Trip Planning

- Trip creation with destination, dates, and budget
- Multi-destination itinerary support
- Day-by-day itinerary builder
- Trip templates for common trip types
- Trip cloning for recurring group trips

Collaboration

- Group trip workspace with member roles
- Real-time collaborative itinerary editing
- Destination and activity voting/polls
- Shared packing lists
- Shared calendars synced to the itinerary
- Task assignment among trip members
- In-app group chat
- Shared travel document folder
- Live travel updates feed for the group

Booking

- Flight search and booking
- Hotel search and booking
- Local transport booking (cabs, transit passes)
- Unified booking confirmation view
- Booking cancellation and modification handling

Expense Management

- Shared expense logging by category
- Automatic expense splitting among members
- Settlement tracking and repayment records
- Budget vs. actual spend tracking
- Multi-currency expense support

Payment

- Integrated payment gateway for bookings
- In-app settlement payments between members
- Payment history and receipts
- Refund handling for cancelled bookings

Notifications

- Web push notifications for booking and itinerary changes
- Reminder notifications for check-ins and deadlines
- In-app notification center

- Notification preference controls

Document Management

- Secure travel document vault (passports, visas, tickets)
- Document expiry tracking and alerts
- Shared access to trip-related documents for group members

Travel Memories

- Photo and note upload tied to itinerary days
- Auto-generated trip photo albums
- AI-assisted travel journal compilation
- Memory sharing with trip members

Analytics

- Personal travel spend and trend dashboard
- Organizer-level trip engagement analytics
- Admin-level platform usage analytics

Admin Dashboard

- User and partner account management
- Content moderation for reviews and memories
- Platform-wide analytics and reporting
- Service health monitoring overview

8. Technology Stack

Layer	Technology	Purpose
Frontend	React.js, TypeScript, Tailwind CSS	Responsive web application UI for travelers and admins
Backend	Node.js, Express.js	REST API services
Architecture	Service-Oriented Architecture (SOA)	Independent, loosely coupled services
APIs	REST APIs, Swagger / OpenAPI	Inter-service and client-service communication and documentation
Authentication	JWT, OAuth	Secure session management and third-party login
Databases	PostgreSQL, MongoDB, Redis	Relational data, flexible document storage, caching
AI	OpenAI API / Gemini API, LangChain (optional)	Itinerary generation, recommendations, assistant
Cloud	Google Cloud Platform, Firebase Storage	Hosting, scaling, and media storage
Maps	Google Maps API	Route display and location-based features
Notifications	Firebase Cloud Messaging	Web push notifications
Payments	Razorpay / Stripe	Payment processing
DevOps	Docker, GitHub Actions	Containerization and CI/CD
Version Control	Git, GitHub	Source control and collaboration
Testing	Jest, Cypress, Postman	Unit, end-to-end, and API testing
Documentation	Swagger, Postman	API documentation and collection sharing
UI/UX	Figma	Interface design and prototyping
Project Management	Jira	Sprint planning and task tracking

9. Roles & Responsibilities

The 8-member team is organized around the platform's service boundaries, which lets members work independently while following a shared API contract.

Role	Team Member Focus	Key Responsibilities
Project Lead & System Architecture	1 member	Overall architecture, service boundaries and API contracts, sprint planning, integration oversight
Authentication & User Service	1 member	Auth Service, User Management Service, role-based access control
AI Module	1 member	AI Travel Assistant, itinerary generation, recommendation and budget-planning logic
Booking Services	1 member	Flight, Hotel, and Local Transport Booking Services and their integrations
Finance & Payment Module	1 member	Budget & Expense Management Service, Payment Service, settlement tracking
Collaboration Module	1 member	Collaborative Itinerary Service, group chat, voting/polls, shared calendars and tasks
Frontend Development (Responsive Web)	1 member	React.js responsive web application covering both traveler and admin views
Testing, Documentation & Deployment	1 member	Test suites, API documentation, CI/CD pipeline, deployment to GCP

Members supporting the AI Module and Booking Services also share responsibility for the Notification, Weather, Currency Conversion, and Recommendation Services, since these are lighter-weight services that plug into the modules above.

10. Timeline & Milestones

Week(s)	Milestone	Key Deliverables
1–2	Requirement Analysis & Literature Review	Finalized requirements, review of existing travel platforms, problem validation
3–4	Architecture & Database Design	Service boundaries, API contracts, ER diagrams, database schemas
5	UI/UX Design	Figma wireframes and prototypes for the responsive web application (traveler and admin views)
6–8	Core Backend Development	Auth, User, and Trip Planning Services implemented and unit-tested
9–10	AI Integration	Itinerary generation, recommendations, and AI assistant integrated
11–12	Booking & Finance Services	Flight, Hotel, Local Transport, Expense, and Payment Services implemented
13	Collaboration Features	Real-time itinerary editing, voting, group chat, shared calendars
14	Frontend Development	Responsive web application (traveler and admin views) connected to backend services
15	Service Integration & System Testing	End-to-end integration testing across all services
16	Deployment & Documentation	Deployment to GCP, finalized API docs, user documentation
17	Final Presentation & Demo	Project demo, final report, and presentation to evaluators